

Why?

Not everything goes according to the plan

Things can fail

- **Network problems**
- **File access problems**
- **User input problems**
- ...

Error handling is crucial

And it is part of your programming job

nil

nil value means that the value is not initialized yet

javascript
null

python
None

java
null

ruby
nil

nil

zero value for pointer-based types

pointers
nil

slices
nil

maps
nil

interfaces
nil

channels
nil

nil

means there is no error

```
err := do()

if err != nil {
    // ❌ error
    // handle it!
    // terminate!
}

// 👍 success
// continue...
```


ERROR VALUE

strconv.Itoa never fails

```
func Itoa(i int) string
```

✓ **this function never returns errors**

ERROR VALUE

strconv.Atoi sometimes **fails**, so you've to handle the **error**

```
func Atoi(s string) (int, error)
```

returns a **value** with an **error type**

✓ **success** → **nil**
(uninitialized error value)

✗ **error** → **non-nil**
(initialized error value)



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nil

```
if err != nil {  
    // ❌ error  
    // quit  
}  
// 👍 success
```


error

```
if err != nil {  
    // ❌ error  
    // quit  
}  
// 👍 success
```


immediately handle the errors

```
if err != nil {  
    // ❌ error  
    // quit  
}  
// 👍 success
```


there is no throw/catch

```
if err != nil {  
    // ❌ error  
    // quit  
}  
// 👍 success
```